Campi Flegrei unrest: provisional observational reproduction and record

Independent reproduction of Tan et al. (2025)

12 October 2026

Main paper

Abstract

Displacement observations, labeled historical source estimates and an external routine earthquake catalog reconstruct the historical context of unrest at Campi Flegrei. Daily GNSS observations now extend through 20 March 2025. Independent waveform processing produces 908 provisional GaMMA associations from 90017 picks, but has not established the publication's relocated fault geometry, magnitude distribution or focal mechanisms. No publication panel is fully independently reproduced. Partial counterparts: 1A, 1B, 1C, S1, S2, S5/map, S5/northing-depth, S5/easting-depth, S8A, S8B, S15C. All remaining panels are not reproduced. The scientific narrative and reproduced supplement precede the input, implementation, validation and completion records.

Campi Flegrei's unrest is recorded by ground displacement and seismicity. Tan et al. (2025) use a machine-learning-assisted earthquake catalog to resolve active structures and evaluate possible sources of unrest. Their study is observational: its central results describe the location, timing and character of earthquakes, rather than solutions of a coupled physical forward model. Here, the completed independent calculations are placed in the same scientific sequence. Published images are confined to the separate comparison PDF.

The current phase of unrest

Figure 1 reconstructs the historical context from the explicitly permitted routine catalog and displacement observations and explicitly labeled historical source estimates. The routine snapshot contains 16,046 located events from 2005 through 20 March 2025, including five events with duration magnitude at least 4.0. Monthly counts include all routine events; the cumulative series includes 7,218 events with $M_d > 0.2$. These external records are separate from the independently detected catalog.

The displacement reconstruction contains 128 historical entries (114 leveling surveys, 13 source-provided estimates and one source-provided reconstruction) and 8,743 daily GNSS observations through 20 March 2025. The original 1905–2023 interval is extended with observed data, and the extension is shaded. A common vertical datum is obtained from overlapping observation records. The long sequence resolves subsidence, the major twentieth-century uplift episodes, and the recent uplift. It does not by itself distinguish magmatic from hydrothermal forcing. All three panels remain partial because the historical catalog snapshot, basemap and exact displacement-stitching convention differ or are incompletely specified.

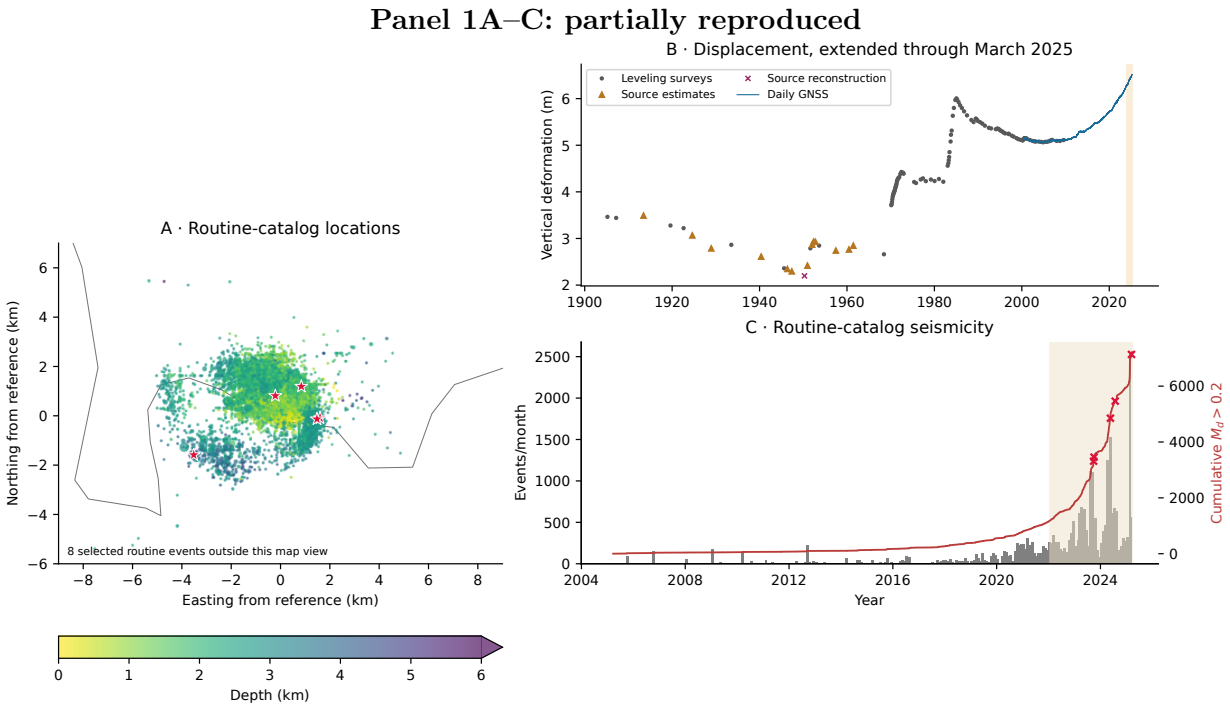


Figure 1: Seismicity and ground displacement. Partial contextual reconstruction. A: external routine catalog, 16,046 located events and five Md>=4 stars. B: 114 leveling surveys (circles), 13 source-provided estimates (triangles), one source-provided reconstruction (cross), and daily GNSS through 20 March 2025; extension beyond 2023 shaded. Historical estimates are not raw observations or reconstructions made here. C: all routine monthly counts and cumulative Md>0.2; March includes days 1–20 only. Routine events are not independent detections. Datum joining is an assumption; basemap and live snapshot differ from the publication.

Improving the earthquake catalog

Independent PhaseNet inference and GaMMA association yield 908 provisional events in the processed waveform sample. These are initial associations with an alternative starting velocity profile. The prior VELEST pilot is quarantined because its background grid is unverified and its fitted velocities are unstable. Regional calibration and double-difference relocation have not established a catalog comparable to Tan et al.'s result. Consequently, they cannot support an independent multiyear seismicity rate or depth–time distribution (Figure 2).

Figure 2 — not reproduced. Panels A and B need multiyear calibrated/relocated events and magnitudes. The current 908 initial associations remain provisional.

A clearer view of the current phase of unrest

Figures 3 and 4 address the publication's main structural observations: the distribution of seismicity and the source mechanisms of larger events. The present independent work has no calibrated, relocated catalog or focal-mechanism solutions. It therefore cannot determine the depth range, continuity or temporal activation of the ring fault, resolve the central fault system, or establish its slip orientations. These unreproduced questions are recorded as text; no placeholder or diagnostic plot is presented as a paper result.

Figure 3 — not reproduced. The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

Figure 4 — not reproduced. Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

Implications for the physical process driving the unrest

The reconstructed hybrid-template example in Figure S15C is an independently processed waveform observation. The limited template search produced 822 unconfirmed threshold crossings; it does not establish recurring hybrid families or their evolution. Neither this result nor the initial associations can test the publication's inference about the presence or absence of upward seismic migration. The partial displacement history also does not distinguish magmatic and hydrothermal forcing. Such an interpretation would require the unresolved spatial and source constraints above.

New perspectives for improved seismic risk assessment

Fault geometry and a calibrated magnitude distribution are needed to connect this analysis to seismic risk. The pilot magnitude measurements have not established the multiyear magnitude-frequency distribution required for Figure S8. The current results therefore provide no revised estimate of event probabilities, damaging-event locations or source dimensions. Their supported contribution is restricted to the partial observational reconstructions shown here and in the supplement.

Reproduced supplement

This supplement follows the original Materials and Methods topic sequence, then Figures S1–S15 and Tables S1–S2. It reports only outcomes supported by the saved independent calculations. Detailed processing settings, input inventories, assumptions and validation are collected after this supplement.

Materials and Methods

S1 Building the Campi Flegrei high-resolution earthquake catalog

Data. The independently retrieved station intervals and station inventory provide partial counterparts to Figures S1 and S2. They do not establish complete network coverage over the study period.

Phase picking. Inference with released trained weights produced 90017 merged picks (13077 P and 76940 S) in the processed observations. This is inference, not reproduction of the training demonstrations S3–S4.

Phase association. GaMMA associated 6526 picks into 908 provisional events using the explicitly labeled published alternative starting profile. These are candidates, not a validated multiyear catalog.

Initial earthquake location. No calibrated location set is accepted. The previous S6 figure and grid-dependent pilot are quarantined. An alternative-profile numerical experiment is documented after the supplement; it is neither S6 nor Table S1. Figure S5 now presents the permitted routine-catalog reconstruction, with 12,503 located study-period events.

Earthquake relocation. Neither TomoDD relocation nor regional HypoDD SVD uncertainty estimation has been completed on an accepted independent catalog. Figures S7 and S9–S13 remain unreproduced.

Local magnitude. The pilot measured 60092 channel amplitudes; 3868 passed the 5 dB SNR threshold. It formed 651 relative-amplitude equations and returned 53 provisional magnitude estimates. Figure S8 displays the partial pilot comparison and distribution. This does not establish the final relocated catalog's magnitude distribution or completeness.

Focal mechanism estimation. SKHASH and gCAP inversions were not run. No independent mechanism is available for Figure 4.

S2 Validation of earthquake locations

Ring-shaped structures with confined depth distributions. The provisional associations do not establish a ring structure. Internal travel-time consistency is preserved in the documentation, but does not substitute for the regional uncertainty or common-receiver tests in Figures S7 and S12.

Depth of earthquakes in the central region. There are now 64 associated CPOZ P arrivals in the provisional sample. P-aligned CPOZ waveform gathers have been saved as a reproduction diagnostic. The relocated sector selections and manual checks on apparently deep central events remain unperformed. No conclusion about a lower depth limit follows from their absence.

S3 Template Matching to Detect Hybrid Events

The search covered 75.797 hours of available common CAAM/COLB intervals and returned 822 unconfirmed threshold crossings. It did not produce independently confirmed families for Figure S14.

Depth. No independent hybrid-event location set is available for Table S2.

Frequency index. Frequency indices were calculated for the retained three templates and the earlier four candidate windows. The expanded scan now has saved eight-second CAAM frequency-index and signal/noise measurements. These use template alignment, not reviewed onsets; coda discrimination and family confirmation remain incomplete. Figure S15C displays the corresponding third written-method template. The first two templates are different examples from the published S15A/B panels and are not inserted in their place.

Relative magnitude. Cluster-relative magnitudes were not assigned because cluster membership has not been established.

Temporal evolution of the hybrid events. The limited exposure and unconfirmed detections do not support a three-year cluster history.

Further search for HB/LP events. The provisional catalog now has station-averaged frequency-index screening on retrieved vertical channels. Missing observations and unreviewed onsets/coda prevent classification of low-frequency candidates or a complete catalog-wide search.

Supplementary figures

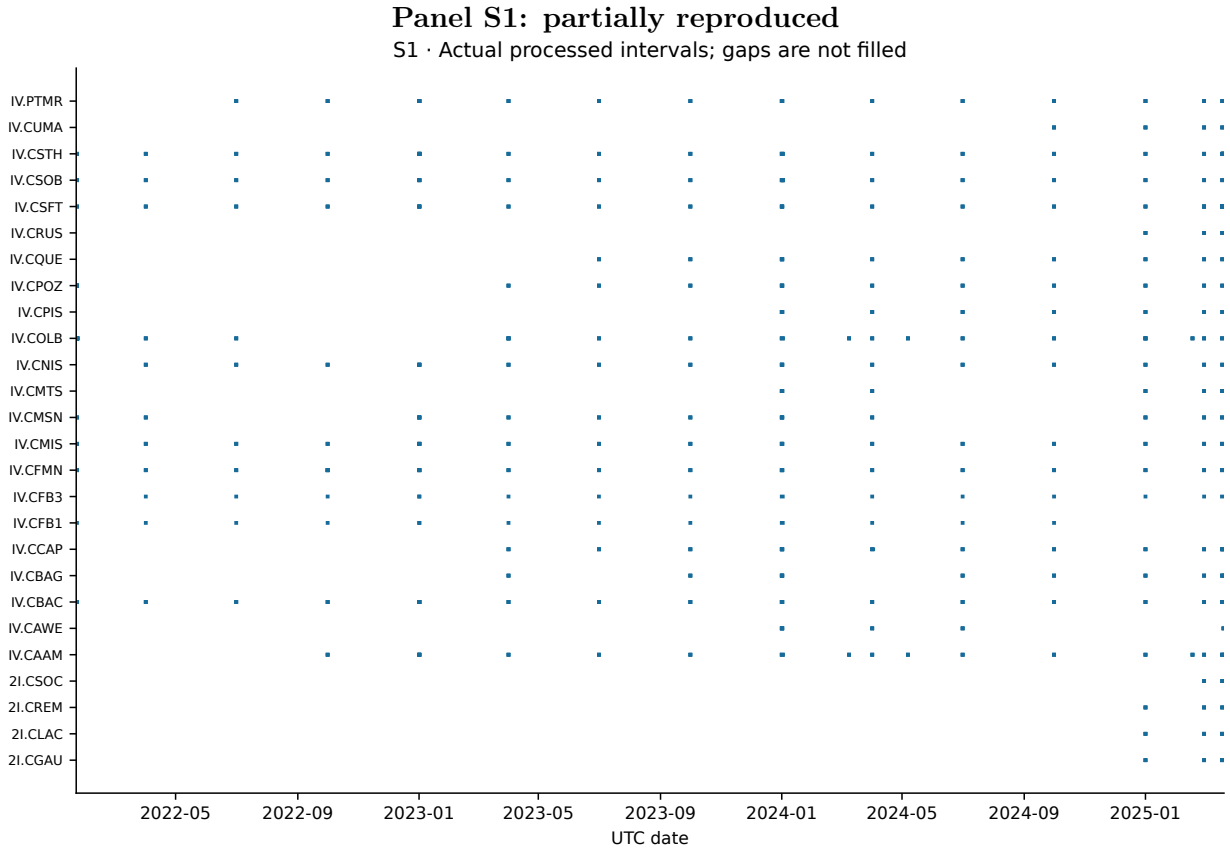


Figure S1: Recorded station availability. Partial: 608 processed files, 26 station names and 4278.694 channel-hours after union of overlapping intervals. Empty intervals are not connected or counted as coverage.

Panel S2: partially reproduced

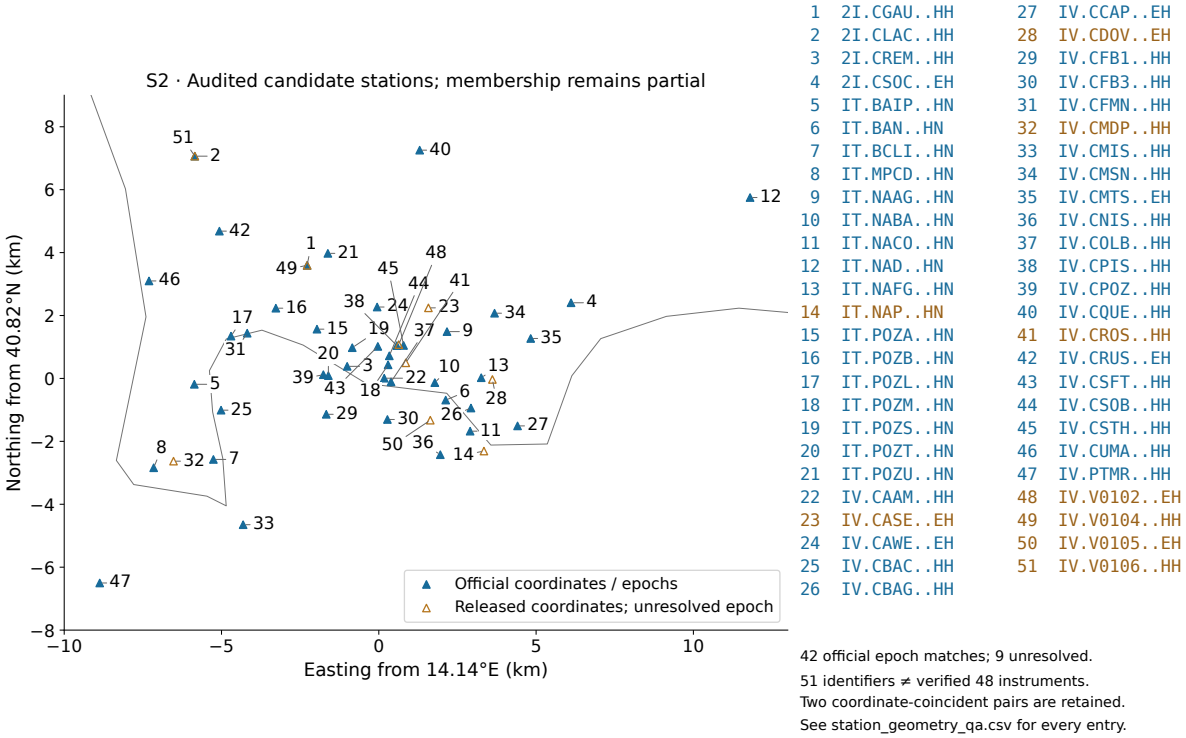


Figure S2: Station locations. Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established.

Figure S3 — not reproduced. Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4 — not reproduced. Panels A and B were not generated for the exact CSOB swarm and CMIS noise examples. Other unlabeled inference comparisons remain diagnostics.

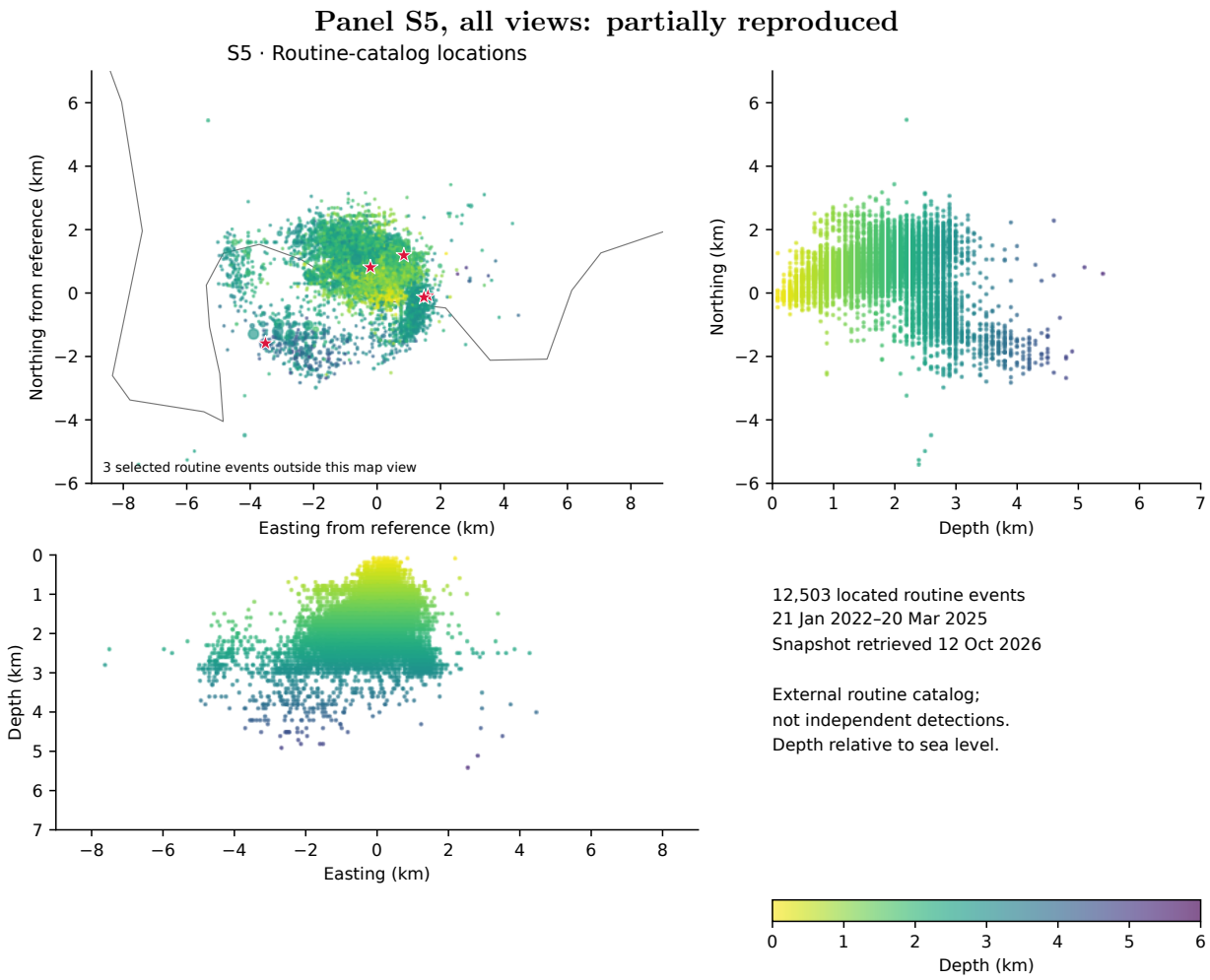


Figure S5: Routine-catalog locations. External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections.

Figure S6 — not reproduced. No accepted current result. For S6, the unverified background grid and dependent previous pilot were quarantined; alternative-profile arithmetic tests are not validated locations. No older output is substituted.

Figure S7 — not reproduced. Panels A–G require independently selected regional subsets, HypoDD SVD locations and coordinate uncertainties. None of these scientific inversions ran.

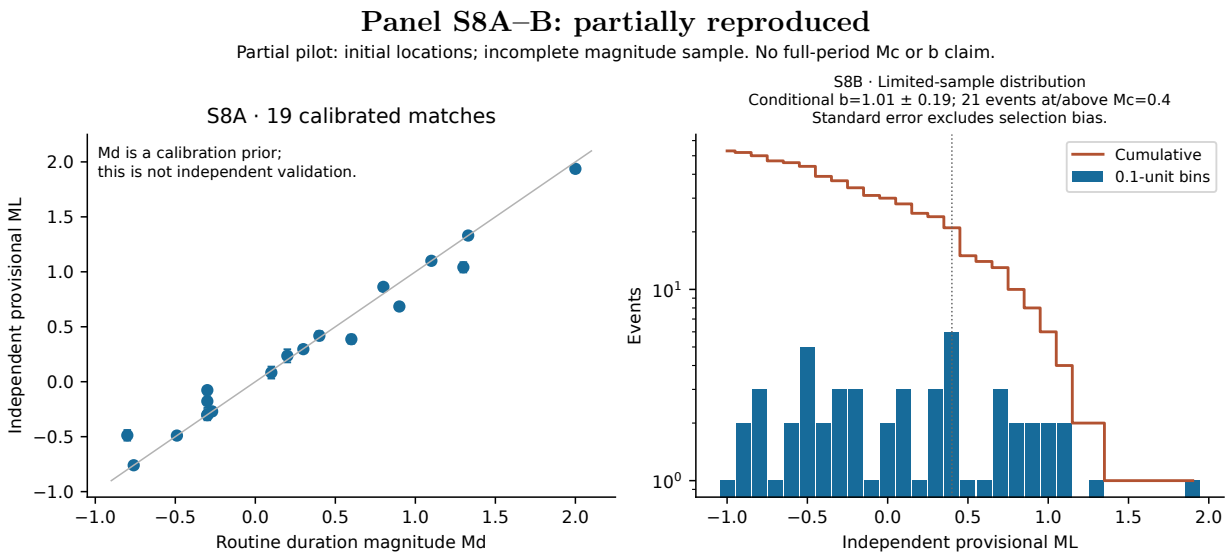


Figure S8: Magnitude calibration and frequency distribution. Partial amplitude-ratio pilot: 53 provisional magnitudes from independent observations, with routine Md as the documented calibration prior. A is not an independent validation of that prior. B is a limited-sample histogram with a conditional likelihood estimate, not the full-period completeness or b-value result. Locations are initial associations, before final relocation.

Figure S9 — not reproduced. All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

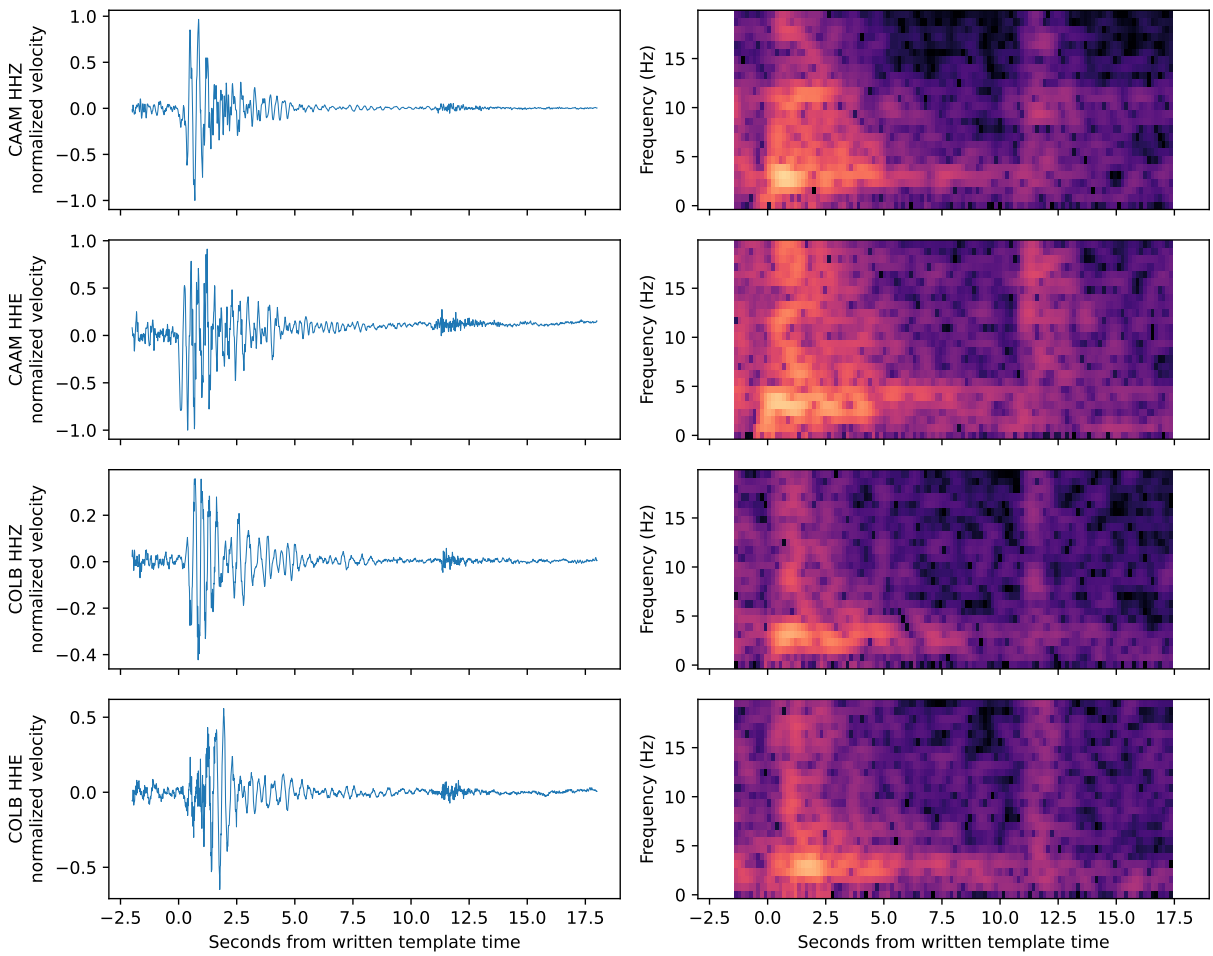
Figure S10 — not reproduced. The four annual groups for 2022–2025 require the independent multiyear relocated catalog. The available sample does not support them.

Figure S11 — not reproduced. The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

Figure S12 — not reproduced. Panels A–H need relocated sector selections and manual depth review. 64 independent CPOZ P arrivals now exist, but the required sector gathers remain unfinished.

Figure S13 — not reproduced. Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

Figure S14 — not reproduced. The cluster histories require three-year exposure, confirmed events and relative magnitudes. The present 75.797 hour scan produced 822 unconfirmed threshold crossings, not reviewed families.



Panels A and B use different published events from the saved E1/E2 templates; D has no independently selected matching example. Those panels remain not reproduced.

Figure S15: Hybrid and shallow-event waveforms. Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

Supplementary tables

Table S1: fitted one-dimensional velocity model — not reproduced. No independently validated regional VELEST model is accepted. Pilot fits and failed attempts, when present, are recorded after the supplement; the author's fitted velocities are not substituted.

Table S2: hybrid events in the relocated catalog — not reproduced. No independently relocated hybrid-event sample exists. The author's event parameters are not substituted.

Supplementary data products

Independent counterparts of the final catalog (Data S1), associated phases of that final catalog (Data S2), focal mechanisms (Data S3), and the relocated interactive view (Interactive Plot S1) are not complete. The saved provisional products retain their existing labels. References and the complete provenance record follow in the reproduction documentation.

Reproduction documentation

This record describes the current continuation, not the superseded nine-event run. Historical outputs remain recoverable from Git. The main paper and reproduced supplement contain only corresponding publication panels; published images occur only in the separate comparison PDF.

D1 Scientific specification and input verification

The local main paper and supplement were read before changing scientific outputs. Main captions 1–4, supplementary captions S1–S15 and the methods are readable. The two uploaded ZIP archives were listed and CRC-tested without executing their contents. Their exact filenames, sizes and checksums remain in the input inventory.

File	Bytes	Inspection
`tanmodel.pdf`	2140554	Text and captions readable
`tanmodelsupp.pdf`	62164947	Text and captions readable
`science.adw9038_data_s1_to_s3.zip`	12960255	Directory readable; CRC passed; not executed
`science.adw9038_interactive_plot_s1.zip`	8852485	Directory readable; CRC passed; not executed

Data S1 is the authors' derived catalog, S2 their associated phases, and S3 their focal mechanisms. The second archive contains an interactive derived-result visualization. None supplies original waveform observations for the independent workflow. These products are excluded from initialization, filtering, calibration, tuning and plot filling. The current workflow does not read these derived tables or compute result-table comparison statistics. Earlier comparison products are quarantined. Published reference images are used only in the separate comparison PDF.

The user now expressly permits the external routine catalog for Figures 1A, 1C and S5, and for the paper's documented duration-magnitude regularization. Those roles are implemented separately from picking and association. Released trained PhaseNet weights, generic method software, original observations and required metadata are permitted. The Tan-fitted 1D model and station corrections, event results, picks and study-specific scripts remain excluded. No figure was digitized or used as numerical input.

D2 Data access and actual coverage

The access refresh found readable miniSEED examples for seven station codes in the temporary archive: CFB1, CFB3, CROS, CRUS, V0102, V0105 and CMDP. CRC, decompression, sample headers and hashes were checked using bounded HTTP ranges; nothing was executed. The saved directory contains 319,889 miniSEED members but no CGAU, CLAC, CREM or CSOC member, although these four names appear on the index page. The examples are access/format checks, not detections or continuous exposure. Their author-event-selected windows and unresolved station epochs prevent treating this archive as an unbiased discovery record. Exact receipts are in `docs/local_waveform_receipts.json`.

Quantity	Value
GNSS observations in plotted interval	8743
GNSS extension in 2025 (days)	79

Quantity	Value
GNSS overlap RMS after offset (m)	0.000295
All historical routine records	26425
Located historical routine records	16046
Located routine study events	12503
Difference from published study count	420
Processed waveform files	608
Processed union channel-hours	4278.694
Merged picks	90017
Provisional GaMMA events	908
Candidates on 19–20 March 2025	829
Successful segmented waveform requests	434
Pending waveform requests	74234

The last two study days account for 829 of 908 provisional events (91.3 percent). This temporal concentration prevents inferring a multiyear seismicity history from this checkpoint. The study interval is 21 January 2022 through 20 March 2025, with the exclusive endpoint 21 March. The previous full-span HTTP 413 responses were request-size failures, not evidence of unavailable observations. The current collector queries FDSN availability and requests six-hour intervals plus half-hour calendar checks with 30-second processing margins. The smaller checks sample the full study date range without implying continuous coverage. Every successful miniSEED response is read, identified by network/station/channel, clipped to its core and hashed. Overlapping channel intervals are merged before computing channel-hours. The original full-period queue contains 74,445 six-hour requests; 229 half-hour calendar checks were added across quarter starts and the study boundaries; unattempted and failed requests never count as coverage. The portable checkpoint and per-request receipts distinguish completed downloads from the pending multiyear workload. At this checkpoint the full period has not been retrieved or processed.

The collector resumes from `inputs/continuation/acquisition.sqlite`, uses six concurrent transfers and stops if shared free space falls below 20 GiB. Linear extrapolation from this download batch gives roughly 970 GB for the entire queue, against about 152 GiB free when checked. That estimate varies with compression and coverage and is not a measured full-archive size. Streaming processing with separate retention or additional storage is needed for the whole queue; pending work is labeled unfinished acquisition, not remote inaccessibility. Processing hours refer to successfully read input channels. Short fragments and missing components can prevent PhaseNet output, and these hours are not claimed as complete three-component detection exposure. The phase-classification warnings are retained in execution logs.

Fresh response-level StationXML was obtained for 22 IV stations and four 2I stations, including historical epochs. Public IT response queries returned HTTP 204. Eight additional IV identifiers missing from the XML—V0106, CDOV, CROS, CASE, V0102, V0105, V0104 and CMDP—were checked individually for full-period availability; all returned HTTP 204. A subsequent all-network StationXML query for these eight codes also returned HTTP 204. These are service-specific no-data responses, not proof that local or author-held records do not exist. RAN's login page explicitly requires registration or credentials to download SAC/ASCII waveforms; those credentials are unavailable. This is an access limitation, not a prompt restriction. Public RAN metadata were subsequently obtained directly from its station directory and XML download: all 17 candidate station files are readable, with 57 study-overlapping HN/HG response epochs. Unknown orientations on ancillary housekeeping channels are not substituted for acceleration-channel orientations. The station directory closes POZA on 15 January 2025, leaving 16 of these stations installed at the study endpoint; this does not establish which instruments entered the paper. Waveform access remains credential-gated. Exact receipts and the archive

listing are in `docs/ran_metadata_access.json` and `docs/ran_metadata_members.json`. The previously accessed local/temporary archive supplies event-selected windows; those windows do not establish continuous exposure and are preserved as such. Exact actual source files are listed in the frozen input and processing manifests. Per-channel response-epoch intersections and every unobserved interval are saved separately; a usable response requires instrument sensitivity and response stages. This verifies metadata availability, not laboratory calibration accuracy. Channel/epoch coordinates differ slightly for some station identifiers; their spread is saved in `coordinate_variability.csv`. Association uses one static station coordinate per identifier, an additional approximation that matters for precise relocation.

The original RITE daily file and format note from Zenodo record 17132077 are readable. They contain MJD, Up displacement and uncertainty in meters, with 8,664 observations through 31 December 2024. No JPG was downloaded as an analysis input. A second INGV release, Zenodo 22259337, published 2 September 2026, contains seven-column observations and format notes through 2025. Its RITE file has 9,025 observations, ending 31 December 2025. We use its 79 daily values through 20 March 2025 to complete the study-period extension. Published MD5 checksums and local SHA-256 hashes were verified. Earlier searches of the Nevada service returned HTTP 404; the public RING RITE position file was readable but ended in 2016 despite its 2025 release timestamp. Neither failure is now treated as a remaining 2025-data blocker.

Routine HTML pages for every year 2005–2025 were retrieved on 12 October 2026. The selected historical interval contains 26,425 records, including unlocated events, and 16,046 located events. The study interval contains 19,274 routine records and 12,503 with latitude, longitude and depth. All five $M_d \geq 4$ events in the Figure 1 description are present. The 420-event excess over the paper's 12,083 located study events is unresolved: archive revisions and undocumented snapshot/selection differences may contribute, but their separate effects cannot be measured without the original historical snapshot. Counts were not tuned. A compressed source snapshot preserves the exact retrieved HTML because live pages can change.

Station geometry and historical source classes

All 51 released station identifiers were compared with official response epochs. Forty-two identifiers have matching epochs; their coordinates replace the rounded released values, with a maximum horizontal correction of 138.8 m. Nine remain unresolved and retain open symbols: eight IV identifiers and NAP, whose official channels are HG rather than the released HN code. The source coordinates are local: the old map scale was enlarged by distant vertices in whole Natural Earth coastline geometries. Those vertices are now masked before projection. Forward/inverse coordinate transforms are checked in kilometers, and the corrected local map retains all candidate identifiers. The released inventory has 25 HH, 9 EH and 17 HN identifiers; the written methods specify 23 broadband, 9 short-period and 16 accelerometer instruments. Two HH pairs share released coordinates, but no alias or deployment record establishes that they represent the same instruments, and the additional HN identifier is unresolved. Thus 51 identifiers cannot yet be reconciled with the reported 48 instruments. The complete QA and epoch tables are `data/continuation/station_geometry_qa.csv` and `data/continuation/station_geometry_epochs.csv`.

The 128 historical entries comprise 114 leveling surveys, 13 source-provided estimates and one source-provided graphical reconstruction. Figure 1B distinguishes them with separate symbols. These estimates were transcribed from the source's numerical table, not extracted from figure pixels; they were not reconstructed by this implementation. Month-only dates are displayed on the fifteenth, with uncertain months retained. The source uncertainty strings and ambiguous original units are preserved without conversion. Daily GNSS observations already cover 2020–2023 and extend through the study endpoint; remaining calendar gaps are unfilled.

D3 Independent processing and explicit departures

Displacement and routine-catalog context

The original Del Gaudio et al. table supplies 128 leveling entries, March 1905–July 2009. Display dates use the middle of the stated month; uncertain months remain flagged. The original March 1905 elevation is 3.464 m. Subtracting it yields a displacement range of −1.266 to 2.541 m; adding the same observed datum back supplies the ordinate used in the new figure. This datum comes from the numerical source table, not from published image pixels. Source notes identify entries estimated historically from photographs or earlier graphical reconstructions; those notes remain in the saved table. This project digitizes no graph.

We align daily GNSS observations to leveling using the median of 23 overlapping epoch offsets, each based on GNSS within 15 days of the leveling date. The offset is 1.6115 m relative to the 1905 displacement zero, with a 0.0025 m median absolute deviation. The 2025 release is aligned using all 8,664 overlapping daily values; its offset is 1.78951 m and overlap RMS after the shift is 0.000295 m. The plotted data contain 319 missing calendar days, which are not filled. Daily continuity is broken across gaps longer than seven days. The present 2025 endpoint is 3.0504 m above the 1905 datum. This assumes comparable displacement at the leveling benchmark and GNSS site; the paper's exact joining recipe is not specified.

Routine selections follow the written Figure 1 and S5 captions. Figure 1A uses located events from 1 January 2005 through 20 March 2025; stars mark $M_d \geq 4$. Figure 1C uses all routine events for monthly counts and strictly $M_d > 0.2$ for the cumulative count. The caption specifies this cutoff only for the cumulative curve. Retaining unlocated records in the monthly bins is therefore an explicit selection assumption; the paper does not specify their treatment, and no filter was chosen to match the published bars. The cumulative curve uses actual event times, avoiding a one-month bin-position shift. March 2025 includes only days 1–20. Figure S5 uses located events from 21 January 2022 onward. There is no additional tuned quality or geographic filter. Coordinates are a WGS84 azimuthal-equidistant projection centered at 14.14°E, 40.82°N; depth is below sea level. Natural Earth coastlines are a display substitute; the original relief and inland zero-elevation contour have not been reconstructed. This is unfinished cartographic work, not a verified access failure. Marker-area scaling is an explicit display choice. The routine catalog never enters the inference or association code.

The contextual map windows are display limits rather than catalog-selection filters. 8 selected historical events lie outside the Figure 1A map window; 3 study-period events lie outside the S5 map and northing–depth view. The easting–depth view omits 0 selected events. These events remain in the saved numerical tables and counts. Axis limits and visible/selected counts are recorded in `data/continuation/routine_plot_scope.csv`; the maps label their off-view counts.

Velocity inputs

The exact Calò–Tramelli (2018) association profile was searched in the readable publisher article and supplement, the article's data statement, INGV's Earth-prints repository, the permitted method-artifact tree, and publications/datasets citing the model. Earth-prints lists only the article PDF in its original-file bundle. Supplement section 2 describes Vp inherited from earlier studies and a depth-dependent Vp/Vs profile revised using Wadati diagrams; Figure S1.2 is graphical, without a numerical layer table. Its data statement directs requests to INGV. No graph was digitized and no Tan-fitted model was substituted.

A numerical alternative is verified in Tramelli et al. (2021), Scientific Reports 11, 7211, Table 1. Layer tops are 0, 0.5, 1, 3 and 10 km below sea level; Vp is respectively 1.7, 2.0, 3.0, 5.5 and 5.7 km/s. Vs is Vp/1.78. Layers are piecewise constant; the top value is extended above sea level for station elevations.

Near-duplicate boundary nodes preserve discontinuities when preparing the 0.1 km eikonal grid. This is a published alternative, not a verified copy of the 2018 starting profile. All affected association/location results remain partial.

The author-distributed `models/3d_vmod` was attributed to Battaglia et al. (2008), but that attribution does not establish numerical identity with the original grid. The official INGV Earth-prints copy of the 2008 paper is readable; its original bundle contains the paper only. The fedOA record has no full text or model file, and the publisher returned HTTP 403. No numerical original grid, checksum or modification record was found. The 2008 written specification gives an 18 by 18 by 9 km volume and 0.5 km spacing. The distributed copy instead has 23 by 14 by 27 nodes on longitude/latitude axes, irregular depth levels and a 200 km boundary node. Its regridding or padding transformations are undocumented here; numerical equivalence cannot be assumed. The downloaded copy, derived regional averages, previous fitted models and S6 locations/figure are quarantined under `quarantine/unverified_background`. They are not eligible inputs to the current workflow. The raw-grid SHA-256 and exact replaced paths remain in its manifest. An independent 2008 grid with provenance, node coordinates, units, Vp/Vs convention and any regridding procedure is needed before exact regional calibration or fixed-grid TomoDD can proceed.

To resolve the exact-profile blocker, request the numerical layer-top depths, Vp and Vs or depth-dependent Vp/Vs underlying Calò–Tramelli supplementary Figure S1.2a/b, both layered and interpolated conventions, units/depth datum, and confirmation of the version used by Tan's GaMMA-1D. The data statement names INGV; Anna Tramelli at INGV–Osservatorio Vesuviano or corresponding author Marco Calò (`calo@igeofisica.unam.mx`) can identify the table. No one was contacted. The full search record is `docs/velocity_profile_search.json`.

Picking, association and calibration

The original released PhaseNet weights are loaded with attribution through SeisBench. Each input is demeaned and high-pass filtered at 1 Hz with a causal four-pole Butterworth filter. Thirty-minute cores have 30-second margins. Both pick thresholds are 0.3; S picks at CASE, CFB1 and CFB3 are excluded. Arrivals within 0.25 s of a higher-probability pick on the same station/phase are merged. Gaps are split, not interpolated. The retraining demonstrations remain unperformed because the exact local annotations, splits and augmentations are unavailable.

GaMMA uses a Bayesian Gaussian mixture, 1D eikonal travel times, at least six picks, and no separate minimum P/S count. Unspecified settings are explicit assumptions: time-only likelihood, DBSCAN epsilon 10 s, minimum samples 3, oversampling factor 5, variance prior 1 s^2, horizontal bounds ± 12 km, and depth 0–8 km. Independent clusters run on six CPU workers and saved identifiers are sorted by origin time. Association is partitioned by UTC day, so phases straddling midnight may be split; this is an additional boundary limitation. All input and coverage margins are clipped to the stated study interval. The first sequential association attempt was restarted with parallel clusters to reduce runtime; neither configuration uses a reference catalog to generate events.

No accepted whole-catalog VELEST solve exists. The current blocker is recorded in `data/continuation/velest_summary.json`; prior dependent products are quarantined. The separate alternative-profile precision experiment is recorded in `data/exploratory/velest_stability/summary.json`. No earlier locations are substituted. The pilot geometry screen requires more than six picks and azimuthal gap below 180°, following generic VELEST guidance; it is not claimed to recreate the paper's manually selected 373 events. Nonpositive or invalid initial propagation times are recorded and removed before inversion. The geometry screen is recomputed; events with fewer than four remaining picks cannot proceed to the next residual-rejection solve. These are explicit numerical departures. Controls use seven iterations, hypocenter/origin damping 0.01, velocity damping

1, station damping 0.1, and S/P observation weight 0.5. The superseded workflow attempted two regional starting models, calibration passes and fixed-model catalog passes with the paper's 0.4 s/0.3 s residual rejection and final six-pick, 0.25 s RMS, 300° gap filters. If a pass fails or produces an unusable result, it is recorded rather than replaced by the authors' fitted solution. Those historical controls, logs and candidate selections are retained in the quarantine audit trail; this grid-dependent sequence is disabled. The current alternative-profile arithmetic experiment does not run the whole-catalog pipeline. Existence of code is not counted as successful inversion.

In the superseded run, 708 candidate events exceeded the generic source's 658-event array capacity. A 1,024-event capacity and bounds checking removed that memory fault; a later 32-bit solve failed a Cholesky check. Those historical failures remain in quarantine. The current isolated experiment compares 32-bit and 64-bit arithmetic from identical observations, the same alternative profile and unchanged damping. It checks both numerical agreement and physical velocity ordering; neither precision is accepted merely because it returns. Current outcomes are reported below. No accepted whole-catalog VELEST solution exists.

The current magnitude pilot includes all retrieved stations with suitable windows, including stations without an associated phase. At those stations it uses theoretical P times from the exploratory alternative profile. Magnitude measurements use picked P arrivals where available, otherwise independent predicted arrivals, a 5 s signal window, a 2 s preceding noise window and a 5 dB power-SNR threshold. Responses are removed to velocity with a stated prefilter and water level. Event separation must be less than one fifth of the common station distances, with at least two common stations. Channel log amplitude ratios are weighted by squared, arrival-aligned zero-lag correlation, an assumption where the precise correlation/weight specification is absent. Weighted least squares uses unit-weight routine-Md priors only for unique post-detection temporal matches within 2 s; multiple independent events claiming the same reference are rejected as ambiguous calibration anchors, without removing detections. Unconnected or unanchored components receive no magnitude. These priors are calibration inputs, not an independent validation set. Provisional initial locations limit any pilot magnitudes. Only full-rank anchored amplitude systems return magnitudes. Formal uncertainties use a residual-scaled inverse normal matrix and remain conditional on the weighting and calibration assumptions. The saved maximum-curvature/maximum-likelihood calculation cannot establish full-period completeness from this noncontinuous sample.

The CAAM/COLB hybrid scan uses the three written-method template times, 8 s templates, 0.2–6 Hz filtering, a two-station mean normalized correlation threshold of 0.25 and 8 s peak separation. Common returned intervals are intersected; duplicates and ten-second filter-edge regions are removed. It covers 75.797 hours and produces 822 unconfirmed threshold crossings. These are not confirmed hybrid clusters. The previous four-candidate review showed persistent fluctuations and did not establish isolated hybrid onsets. The extended candidates now receive saved CAAM frequency-index and signal/noise measurements; these measurements alone do not confirm events. Cluster membership, three-year history, cluster-relative magnitudes and catalog-wide coda discrimination remain unfinished.

D4 Supporting results and validation

The current run has 90017 merged picks and 908 provisional associations. Probability bounds, S-pick exclusions, the six-pick minimum, unique assignment and valid pick indices are checked. Synthetic homogeneous-medium, analytic and injected-signal fixtures are isolated under `tests/results`. They test software assertions only and are excluded from empirical catalogs, magnitude measurements and paper figures. The rebuild reruns the equation and layered-ray checks; their pass records are not observations.

The present S–P check includes 1405 paired observations, of which 57 are nonpositive. There are

64 associated CPOZ P arrivals. These defects and the alternative velocity profile prevent treating raw associations as validated locations. Internal residuals from the same model do not establish absolute accuracy. The location-check tables retain failed observations. The previous generic/retrained comparison on an unlabeled 1600 s CMIS window produced 17 P/4 S and 11 P/55 S picks respectively; the 120 s CSOB window produced zero and one pick. Those counts remain supporting evidence only, not training or labeled-accuracy reproduction, and are not inserted in S4. An analytic vertical-ray check of the published alternative gives maximum P/S errors of 0.040642/0.072342 s on the 0.1 km eikonal grid; halving the spacing reduces them to 0.020321/0.036171 s. This confirms first-order interface discretization behavior, not the physical accuracy of the alternative profile. The numerical checks and refinement results are saved in `tests/results/profile_ray_checks.csv`.

The current amplitude/SNR stage measured 60092 channel amplitudes, with 3868 passing SNR and 651 accepted difference equations. It returned 53 provisional magnitudes. These are actual executed outputs, not a promised future count. Candidate measurement errors and quality failures are retained in the JSON/CSV records. The old 140-measurement, 23-SNR, zero-edge result is superseded for the current sample and preserved in Git and the quarantine audit trail.

Additional observational checks

The CPOZ diagnostic contains 64 P-aligned observed waveforms. It does not implement the relocated ring-sector test.

Spectral measurements were calculated for 822 of 822 template crossings. None is confirmed as a hybrid event; spectral windows use correlation alignment and still require onset/coda review.

Catalog frequency-index screening measured 20631 station windows for 908 provisional events. Station FIs are averaged; no low-frequency candidate is classified as a confirmed hybrid without onset and coda review.

The alternative-profile VELEST experiment used 120 provisional events. Both real precisions were attempted with identical serialized inputs.

Maximum 32/64-bit differences were 0.000 km/s for Vp and 0.000 km/s for Vs, with 0.000 km in fitted depth.

Both runs pass the necessary positive-Vp-greater-than-Vs check. This ordering and numerical agreement do not replace reviewed phases or the documented regional calibration; no experimental fit is accepted for S6 or relocation.

The 32-bit experiment has event-RMS aggregate 0.389 s.

The 64-bit experiment has event-RMS aggregate 0.389 s.

Magnitude statistics and comparison boundary

The conditional magnitude-frequency fit gives $M_c = 0.4$ and $b = 1.01 \pm 0.19$ from only 21 selected magnitudes at or above the pilot completeness level. The quoted Shi–Bolt standard error excludes coverage, selection and magnitude-calibration biases. This calculation does not establish population completeness or a regional b-value.

No Tan-derived result-table statistics are used in this paper. The comparison PDF contains only clearly labeled published figure references and the eligible independent counterparts.

D5 Limitations and unfinished analyses

No publication panel is fully independently reproduced. Partial counterparts: 1A, 1B, 1C, S1, S2, S5/map, S5/northing-depth, S5/easting-depth, S8A, S8B, S15C. All remaining panels are not reproduced. Routine-context permissions and the 2025 displacement gap are resolved. The full continuous download/processing workload remains unfinished. Exact Calò–Tramelli layer values are an external-input limitation; the released fixed 3D artifact is quarantined because original numerical provenance is unresolved. RAN credentials are a specific access limitation.

The regional calibrated and TomoDD-relocated catalog, four HypoDD SVD subsets, reliable multi-year magnitudes/completeness, manually reviewed polarities and S/P ratios, SKHASH/gCAP solutions, FK Green's functions, exact event-specific gCAP filters/windows/weights, relocated CPOZ sector gathers, manual deep-event review and three-year hybrid clustering are not complete. Generic TomoDD source has now been retrieved, classified and compiled; compilation alone is not relocation. Training labels/splits and some inversion settings are incompletely specified. Supplied author-derived results could fill plots only by violating the independence restriction, so they are not substituted. That restriction is distinct from unfinished implementation or genuinely missing source inputs.

D6 Figure-by-figure record

The machine-readable `docs/panel_record.json` records every panel's written specification, parameters, exact input paths, executed data/plot commands, saved products and limitation. The raw-waveform manifest is `docs/continuation_processed_inputs.json`; its intervals are a frozen checkpoint, not the full queue. Context panels use `continue_context.py` and `plot_continuation.py`. S1 uses `process_continuous.py` then `plot_continuation.py`. S2 is audited by `scripts/station_geometry.py`; S15C retains its original-waveform recipe. Unmatched panels have text notices in this paper and explicit placeholders only in the comparison PDF.

Panel/view	Quantity	Status, specification and limitation
1A	INGV event map and large-event markers	partially reproduced. Main p.71, Fig.1 caption. 2005-01-01 to 2025-03-20; origin 14.14 E, 40.82 N; sea-level depth; Md>=4 markers. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline is a display substitute; the original relief/inland-zero contour has not been reconstructed.
1B	Vertical displacement at Pozzuoli port	partially reproduced. Main p.71, Fig.1 caption. 128 historical entries: 114 leveling surveys, 13 source-provided estimates and one source-provided graphical reconstruction. Source classes preserved separately. Daily GNSS 2000-05-29 through 2025-03-20; median datum from 23 overlapping leveling epochs, +/-15 days. Month-only dates placed on day 15 for display.. Benchmark equivalence and datum-joining convention are assumed because the exact original stitching recipe is not specified. Calendar gaps remain unfilled; no missing recent-data blocker through 2025-03-20.
1C	Temporal seismicity and cumulative counts	partially reproduced. Main p.71, Fig.1 caption. 2005-2025; cumulative Md>0.2; study interval 2022-01-21 to 2025-03-20. Monthly counts and exact-time cumulative Md>0.2 reconstructed from the current routine snapshot; historical count differences unresolved. March 2025 is a partial month.

Panel/view	Quantity	Status, specification and limitation
2A	Catalog seismicity rates by region	not reproduced. Main p.72, Fig.2 caption. Central vs ring/other regions; 54319 vs 12083 catalog events; time bin unspecified. 908 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete.
2B	Depth versus time, symbol size by magnitude	not reproduced. Main p.72, Fig.2 caption. Sea-level depth; near-surface 0-0.5 km; central 0.5-2.1; ring sectors to 3.7 km. 908 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete.
3/map	Epicentral spatial pattern	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. 908 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete.
3/northing-depth	Depth versus northing	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. 908 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete.
3/ easting-depth	Depth versus easting	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. 908 provisional GaMMA events are not a complete calibrated, TomoDD-relocated catalog. Multiyear acquisition and downstream processing remain incomplete.
4A	Ring-sector focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Accepted independent locations/model, reviewed P polarities and S/P ratios, Green functions and event-specific gCAP settings are missing or unfinished. SKHASH/gCAP not run. The written methods and figure caption use different counts; no supplied mechanism table is used to resolve this in the scientific report.
4B	Central/onshore focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Accepted independent locations/model, reviewed P polarities and S/P ratios, Green functions and event-specific gCAP settings are missing or unfinished. SKHASH/gCAP not run. The written methods and figure caption use different counts; no supplied mechanism table is used to resolve this in the scientific report.
S1	Station data-availability timeline	partially reproduced. Supplement S12 caption; S2 methods. 48 stations; continuous-data availability. 608 files processed, 4278.694 union channel-hours. Full-period queue still pending; actual gaps and unavailable RAN credentials remain. Requests/availability are not coverage.
S2	Station map	partially reproduced. Supplement S13 caption; S2 methods. 23 broadband, 9 short-period, 16 accelerometers. The published 48-instrument membership is not verified. Two released-coordinate pairs coincide; no deployment alias mapping justifies collapsing or removing identifiers.
S3A	Noise example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.

Panel/view	Quantity	Status, specification and limitation
S3B	Single-event example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3C	Two overlapping events: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3D	Five overlapping copies of one event: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S4A	Swarms at CSOB: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 120 s; 2 h after May 2024 Md4.4 event; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark.
S4B	CMIS noise: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 1600 s; coastal noise; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark.
S5/map	Routine-catalog spatial views; map	partially reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline is a display substitute; the original relief/inland-zero contour has not been reconstructed.
S5/northing-depth	Routine-catalog spatial views; northing-depth	partially reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline is a display substitute; the original relief/inland-zero contour has not been reconstructed.
S5/ easting-depth	Routine-catalog spatial views; easting-depth	partially reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Current catalog snapshot differs from the historical snapshot: 12,503 located study events versus 12,083. No tuning to published counts. Natural Earth coastline is a display substitute; the original relief/inland-zero contour has not been reconstructed.
S6/map	Initial-location spatial views; map	not reproduced. Supplement S17 caption. Requires verified fixed 2008 grid, documented regional averages and validated independent VELEST inversions.. Fixed 3D grid provenance remains unverified; previous precision-sensitive regional calibration is rejected. The exploratory 2021-profile experiment is not an accepted regional model. No old or author locations substituted.

Panel/view	Quantity	Status, specification and limitation
S6/northing-depth	Initial-location spatial views; northing-depth	not reproduced. Supplement S17 caption. Requires verified fixed 2008 grid, documented regional averages and validated independent VELEST inversions.. Fixed 3D grid provenance remains unverified; previous precision-sensitive regional calibration is rejected. The exploratory 2021-profile experiment is not an accepted regional model. No old or author locations substituted.
S6/ easting-depth	Initial-location spatial views; easting-depth	not reproduced. Supplement S17 caption. Requires verified fixed 2008 grid, documented regional averages and validated independent VELEST inversions.. Fixed 3D grid provenance remains unverified; previous precision-sensitive regional calibration is rejected. The exploratory 2021-profile experiment is not an accepted regional model. No old or author locations substituted.
S7A/map	Cluster locations and 2-sigma errors: map and cross-sectional views; map	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7A/ easting-depth	Cluster locations and 2-sigma errors: map and cross-sectional views; easting-depth	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7B	EX uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7D	EY uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7F	EZ uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7C	EX uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7E	EY uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7G	EZ uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.

Panel/view	Quantity	Status, specification and limitation
S8A	Estimated local versus routine duration magnitude	partially reproduced. Supplement S19 caption; S5-S6 methods. 5 s signal, 2 s preceding noise; SNR >=5 dB; all available recorded stations; own picked or predicted P arrivals; >=2 shared stations; separation < one fifth of station distance; squared zero-lag correlation weights; Md regularization lambda=1 for unique one-to-one matches within 2 s. Initial locations. B uses 0.1 magnitude bins, maximum-curvature Mc, half-bin-corrected likelihood b and formal Shi-Bolt sampling error; no arbitrary minimum sample cutoff.. 60092 channel measurements, 3868 passing SNR, 651 relative-amplitude equations, 53 provisional magnitudes. Final relocated locations/full exposure remain missing; routine Md is an allowed regularizer, not independent validation.
S8B	Magnitude-frequency distribution and fitted b value	partially reproduced. Supplement S19 caption; S6 methods. 5 s signal, 2 s preceding noise; SNR >=5 dB; all available recorded stations; own picked or predicted P arrivals; >=2 shared stations; separation < one fifth of station distance; squared zero-lag correlation weights; Md regularization lambda=1 for unique one-to-one matches within 2 s. Initial locations. B uses 0.1 magnitude bins, maximum-curvature Mc, half-bin-corrected likelihood b and formal Shi-Bolt sampling error; no arbitrary minimum sample cutoff.. 60092 channel measurements, 3868 passing SNR, 651 relative-amplitude equations, 53 provisional magnitudes. Final relocated locations/full exposure remain missing; routine Md is an allowed regularizer, not independent validation.
S9/map	Filtered-catalog spatial views and feature labels A-D; map	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S9/northing-depth	Filtered-catalog spatial views and feature labels A-D; northing-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S9/ easting-depth	Filtered-catalog spatial views and feature labels A-D; easting-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2022	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2022. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2023	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2023. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2024	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2024. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.

Panel/view	Quantity	Status, specification and limitation
S10/2025	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2025 through March20. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-1	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-2	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-3	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-4	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S12A	Ring-sector division	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S12B	Ring-sector division in complementary spatial view	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S12C	Common-receiver gather: east / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S12D	Common-receiver gather: southwest / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S12E	Common-receiver gather: north / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.

Panel/view	Quantity	Status, specification and limitation
S12F	Common-receiver gather: east / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S12G	Common-receiver gather: southwest / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S12H	Common-receiver gather: north / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. 64 independently associated CPOZ P arrivals now exist. Relocated ring-sector selection and manual depth review remain unfinished; internal S-P checks do not establish location accuracy.
S13A	Full caldera seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S13B	Onshore seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S14/1	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 75.797 common CAAM/COLB scan hours; 822 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished.
S14/2	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 75.797 common CAAM/COLB scan hours; 822 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished.
S14/3	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 75.797 common CAAM/COLB scan hours; 822 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished.
S14/ unclustered	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. 75.797 common CAAM/COLB scan hours; 822 unconfirmed threshold crossings. Three-year scan, reviewed clusters, relative magnitudes and timing histories remain unfinished.
S15A	Hybrid family 1: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.

Panel/view	Quantity	Status, specification and limitation
S15B	Hybrid family 2: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.
S15C	Hybrid family 3: waveforms and spectrograms	partially reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Written template times used; exact published example alignment, display filtering and spectral settings incompletely specified. Sensitivity-corrected velocities and stated independent display settings used.
S15D	Shallow volcano-tectonic example: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Exact shallow VT example not identified by written method alone; no author catalog or plot values used to select it.

Executed partial-panel chains

The following commands generate the saved numerical files and their plotted counterparts. The full input lists and checksums are retained in the machine-readable records; no unreproduced panel has an executed numerical counterpart. The source-to-output checks cover the explicit current graph and do not prove the absence of unexamined or hidden inputs.

1A — partial data-to-plot chain. External routine-catalog context; not independent detections.

21 exact input files are listed under this panel in `docs/panel_record.json`; waveform hashes and intervals are in `docs/continuation_processed_inputs.json`, with source receipts in the input manifests.

Data command:

```
.venv/bin/pythonscripts/continue_context.py
```

Saved numerical output: `data/continuation/figure_1A.csv`.

Plot command: `.venv/bin/pythonscripts/plot_continuation.py`

Output: `figures/figure_1A_routine.pdf`.

The same plotting command composes the report figure: `figures/figure_1_combined.pdf`.

1B — partial data-to-plot chain. Original GNSS/leveling observations plus explicitly labeled historical source estimates/reconstruction.

Inputs: `inputs/continuation/RITE_2000_2025.txt`, `inputs/continuation/RITE_Timeseries_Format.readme`, `inputs/continuation/RITE_daily.txt`, `inputs/continuation/Time_Series_Format.txt`, `inputs/extended/delgaudio2010_table.doc`.

Data command:

```
.venv/bin/pythonscripts/continue_context.py
```

Saved numerical output: `data/continuation/rite_gnss.csv`, `data/continuation/historical_leveling.csv`.

Plot command: `.venv/bin/pythonscripts/plot_continuation.py`

Output: `figures/figure_1B_displacement.pdf`.

The same plotting command composes the report figure: `figures/figure_1_combined.pdf`.

1C — partial data-to-plot chain. External routine-catalog context; not independent detections.

21 exact input files are listed under this panel in `docs/panel_record.json`; waveform hashes and intervals are in `docs/continuation_processed_inputs.json`, with source receipts in the input manifests.

Data command:

.venv/bin/pythonscripts/continue_context.py

Saved numerical output: data/continuation/figure_1C.csv, data/continuation/figure_1C_cumulative.csv.

Plot command: .venv/bin/pythonscripts/plot_continuation.py

Output: figures/figure_1C_routine.pdf.

The same plotting command composes the report figure: figures/figure_1_combined.pdf.

S1 — partial data-to-plot chain. Own processing of observations/metadata; trained weights and external Md calibration are identified in the method record.

608 exact input files are listed under this panel in docs/panel_record.json; waveform hashes and intervals are in docs/continuation_processed_inputs.json, with source receipts in the input manifests.

Data command:

.venv/bin/pythonscripts/process_continuous.pyall--frozen

Saved numerical output: data/continuation/processed_coverage_union.json.

Plot command: .venv/bin/pythonscripts/plot_continuation.py

Output: figures/continuation_coverage.pdf.

S2 — partial data-to-plot chain. Own processing of observations/metadata; trained weights and external Md calibration are identified in the method record.

22 exact input files are listed under this panel in docs/panel_record.json; waveform hashes and intervals are in docs/continuation_processed_inputs.json, with source receipts in the input manifests.

Data command:

.venv/bin/pythonscripts/station_geometry.py

Saved numerical output: data/continuation/figure_S2.csv, data/continuation/station_geometry_qa.csv, data/continuation/station_geometry_epochs.csv.

Plot command: .venv/bin/pythonscripts/plot_continuation.py

Output: figures/continuation_stations.pdf.

S5/map — partial data-to-plot chain. External routine-catalog context; not independent detections.

21 exact input files are listed under this panel in docs/panel_record.json; waveform hashes and intervals are in docs/continuation_processed_inputs.json, with source receipts in the input manifests.

Data command:

.venv/bin/pythonscripts/continue_context.py

Saved numerical output: data/continuation/figure_S5.csv.

Plot command: .venv/bin/pythonscripts/plot_continuation.py

Output: figures/figure_S5_routine.pdf.

S5/northing-depth — partial data-to-plot chain. External routine-catalog context; not independent detections.

21 exact input files are listed under this panel in docs/panel_record.json; waveform hashes and intervals are in docs/continuation_processed_inputs.json, with source receipts in the input manifests.

Data command:

.venv/bin/pythonscripts/continue_context.py

Saved numerical output: data/continuation/figure_S5.csv.

Plot command: .venv/bin/pythonscripts/plot_continuation.py

Output: figures/figure_S5_routine.pdf.

S5/easting-depth — partial data-to-plot chain. External routine-catalog context; not independent detections.

21 exact input files are listed under this panel in docs/panel_record.json; waveform hashes and intervals are in docs/continuation_processed_inputs.json, with source receipts in the input manifests.

Data command:

`.venv/bin/pythonscripts/continue_context.py`

Saved numerical output: `data/continuation/figure_S5.csv`.

Plot command: `.venv/bin/pythonscripts/plot_continuation.py`

Output: `figures/figure_S5_routine.pdf`.

S8A — partial data-to-plot chain. Own processing of observations/metadata; trained weights and external Md calibration are identified in the method record.

631 exact input files are listed under this panel in `docs/panel_record.json`; waveform hashes and intervals are in `docs/continuation_processed_inputs.json`, with source receipts in the input manifests.

Data command:

`.venv/bin/pythonscripts/downstream_checks.py--datasetcontinuation`

`.venv/bin/pythonscripts/magnitude_statistics.py`

Saved numerical output: `data/continuation/figure_S8A.csv`.

Plot command: `.venv/bin/pythonscripts/plot_continuation.py`

Output: `figures/figure_S8_pilot.pdf`.

S8B — partial data-to-plot chain. Own processing of observations/metadata; trained weights and external Md calibration are identified in the method record.

631 exact input files are listed under this panel in `docs/panel_record.json`; waveform hashes and intervals are in `docs/continuation_processed_inputs.json`, with source receipts in the input manifests.

Data command:

`.venv/bin/pythonscripts/downstream_checks.py--datasetcontinuation`

`.venv/bin/pythonscripts/magnitude_statistics.py`

Saved numerical output: `data/continuation/figure_S8B.csv`, `data/continuation/magnitude_statistics.json`.

Plot command: `.venv/bin/pythonscripts/plot_continuation.py`

Output: `figures/figure_S8_pilot.pdf`.

S15C — partial data-to-plot chain. Own processing of observations/metadata; trained weights and external Md calibration are identified in the method record.

Inputs: `inputs/metadata/response.xml`, `inputs/waveforms/E3_CAAM.mseed`, `inputs/waveforms/E3_COLB.mseed`.

Data command:

`.venv/bin/pythonscripts/analyze_observations.py`

Saved numerical output: `data/observations/E3_waveforms.csv`, `data/observations/E3_CAAM_Z_spectrum.npz`, `data/observations/E3_CAAM_E_spectrum.npz`, `data/observations/E3_COLB_Z_spectrum.npz`, `data/observations/E3_COLB_E_spectrum.npz`.

Plot command: `.venv/bin/pythonscripts/plot_observations.py`

Output: `figures/hybrid_E3.pdf`.

D7 Saved numerical tables and equation consistency checks

Original-waveform template frequency indices

These retained numerical measurements describe written-method templates, not confirmed hybrid families. E1 and E2 are not substituted for the different published S15A/B examples.

Template	FI	Samples
E1	-0.7862	800

Template	FI	Samples
E2	-0.4286	800
E3	-1.0819	800

Rejected velocity experiments

The prior grid-dependent regional calibration is archived, not accepted. The separate `scripts/velest_stability.py` experiment starts from the cited Tramelli (2021) numerical table. It runs identical serialized observations and damping with 32-bit and 64-bit real arithmetic. Raw model outputs, fitted-location files and physical-velocity checks are saved under `data/exploratory/velest_stability`. Even precision agreement cannot establish a correct physical model; any $V_p \leq V_s$ layer rejects the fit. These results do not supply S6, Table S1, or downstream relocation. The following unaccepted experimental values come directly from `model_32.csv` and `model_64.csv` in that directory, generated by `.venv/bin/pythonscripts/velest_stability.py`. Velocities are in km/s. This is a reproduction diagnostic table, not a counterpart to Table S1.

Layer top (km b.s.l.)	V_p, 32 bit	V_s, 32 bit	V_p, 64 bit	V_s, 64 bit
-0.50	1.32	0.88	1.32	0.88
0.50	2.42	1.27	2.42	1.27
1.00	2.51	1.57	2.51	1.57
3.00	5.20	3.26	5.20	3.26
10.00	5.70	3.20	5.70	3.20

D8 Rebuild and replacements

The amplitude stage uses sixteen spawned workers and bounded caches of unchanged input waveforms; each measurement window is copied before processing. A software check compares 60 observed windows against direct reads and requires identical samples and timestamps. An earlier amplitude run was intentionally interrupted to avoid repeated decoding. This performance restart was not an access failure; completed inference and association were retained only after their recorded hashes were verified. The following measurement pass reached the inversion-output step but failed when JSON serialization encountered a NumPy Boolean. The acceptance flag was converted explicitly to a Python Boolean, and the full failed stage was rerun from observations. The failed attempt's saved numerical hashes are preserved in `docs/magnitude_serialization_failure.json`; it is not counted as a passed build. Six downstream CSVs, including the magnitude equations and estimates, are byte-identical between that failed pass and the successful rerun. The amplitude CSV hash differs. Because its pre-failure bytes were not retained, the numerical or ordering cause of that difference cannot be resolved from the hash alone; no claim of byte-for-byte amplitude reproducibility is made. The current measurements come from the complete passed rerun. The clean continuation procedure is documented in `docs/CONTINUATION.md`. Restore the frozen inputs with `scripts/restore_inputs.py--continuation`; run `scripts/rebuild_continuation.py--fresh` using the repository's Python environment. The rebuild regenerates observations, picking/association, the downstream attempts, context panels and both PDFs. It is a rebuild of the frozen observed subset, not proof of complete multiyear reproduction. Its actual outcome and executed commands are saved in `data/continuation/rebuild_status.json` and `data/continuation/rebuild_log.json`. The current fresh rebuild first stopped because this Python installation lacks SQLite's in-memory deserialization method. The ledger reader was changed to a temporary project-local database; missing request segment

fields also required explicit null handling. Passed stages were hash-verified before resumption. Earlier worker and study-boundary failures belong to superseded runs and are not evidence of unavailable observations. The document build verifies main-text sections, the eligible generated figures in original numbering order, supplement topic order, and all nineteen comparison pages in original order, and confirms that published images occur only in the comparison PDF.

This continuation replaces the prior report/comparison PDFs and generated figure fragments, panel records and obsolete context figure usage. The old displacement display ending in 2019 is replaced by observed daily coverage through March 2025. The old context plots are superseded by the corresponding Figure 1 and S5 outputs; their scripts/data remain traceable, while their obsolete PDFs/PNGs are removed after the current counterparts exist. New seismic outputs are in `data/continuation`. Eligible baseline observations and picks remain in `data/extended`; grid-dependent baseline results and obsolete comparison statistics are in `quarantine`. Synthetic fixtures are isolated in `tests/results`. Exact changes are documented in `docs/REPLACEMENTS.md`. Project-specific caches remain under `tmp`; pip's normal cache is `/home/mehffin/.cache/pip`. Large waveforms and model/software artifacts remain ignored under `inputs`, with retrieval URLs, file sizes and hashes.

D9 Sources

The primary specification is Tan et al. (2025), Science 390, 70–75, local `tanmodel.pdf` and `tanmodelsupp.pdf`. Observation/metadata services: https://webservices.ingv.it/fdsnws/; public RAN station metadata: https://ran.protezionecivile.it/IT/stazioni.php; temporary waveform archive: https://dataserver.ov.ingv.it/users/anna.tramelli/seismic_records/index.html; routine catalog: https://terremoti.ov.ingv.it/gossip/flegrei/. GNSS: De Martino and colleagues, https://zenodo.org/records/17132077 and https://zenodo.org/records/22259337, with the requested citation to De Martino et al. (2021), doi:10.3390/rs13142725. Historical leveling: Del Gaudio et al. (2010), doi:10.1016/j.jvolgeores.2010.05.014. Published alternative profile: Tramelli et al. (2021), doi:10.1038/s41598-021-86506-6, Table 1. Requested exact profile: Calò and Tramelli (2018), doi:10.1038/s41598-018-34456-x. Quarantined, unverified grid attribution: Battaglia et al. (2008), doi:10.1111/j.1365-2478.2007.00687.x.

Tan and colleagues' released trained PhaseNet artifact is attributed to https://github.com/iamharrison2000/Campi-Flegrei-Earthquake-Monitoring. Only classified weights, configuration and station metadata enter the current workflow. The permitted artifact revision is

`f84c9c9dc6bb1501a95a833f6634199a63d4f3b3`.

GaMMA revision is

`80394dd4a4c29450da599f507916ecf1db6203dd`.

Generic VELEST, HypoDD and TomoDD source receipts remain in the repository. No Tan-specific analysis or plotting script was used.